

TESTING EVIDENCE

Automated test results

```
> unit test/unit/auth-password.test.ts [queued]

Test Files 0 passed (3)
Tests 0 passed (0)

> unit test/unit/auth-lockout.test.ts [queued]
> unit test/unit/auth-password.test.ts [queued]
> unit test/unit/example.test.ts [queued]

Test Files 0 passed (3)
Tests 0 passed (0)
✓ unit test/unit/example.test.ts (1 test) 6ms
✓ unit test/unit/auth-password.test.ts (9 tests) 11ms
✓ unit test/unit/auth-lockout.test.ts (8 tests) 15ms

Test Files 3 passed (3)
Tests 18 passed (18)
Start at 01:44:35
Duration 1.03s (transform 369ms, setup 0ms, import 941ms, tests 32ms, environment 1ms)
```

Test case 1 - Guest journey: search, book, view invoice

This walkthrough proves a paying guest can find a room, attach optional services, see a live total, and end up with a printable invoice.

Log in at login using john.smith@gmail.com / Guest123!

Right after login, you land on the dashboard

Welcome back, John

Manage your reservations and plan your next stay.

[Search rooms](#) [All bookings](#) [Profile](#)

ACTIVE STAY 0	UPCOMING 2	PAST STAYS 1	TOTAL SPENT £420.00
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Upcoming bookings

[View all →](#)

#	HOTEL	CHECK-IN	CHECK-OUT	TOTAL	
2	Verdan Paris	25 May 2026	28 May 2026	£600.00	Details
3	Verdan New York	20 Jul 2026	25 Jul 2026	£3750.00	Details

Recent history

[View all →](#)

#	HOTEL	DATES	STATUS	TOTAL	
1	Verdan London	9 Apr 2026 → 12 Apr 2026	CHECKED_OUT	£420.00	Invoice →

Guests are redirected to a personalised dashboard immediately on login. The stat cards summarise their active stay, upcoming bookings, past stays and total spend, while the table previews their next reservation. This shows the role-based redirect is working — guests do not see staff/admin views.

Click *Search rooms* in the welcome banner. Enter check-in = 2026-07-10 and check-out = 2026-07-13, then click Search.

Verdan Hotels

Search room availability across our London, Paris, and New York properties.

 My dashboard

 My bookings

Find a room

City

Any city

Check-in

07/10/2026

Check-out

07/13/2026

Guests

Any

Room type

Any type

Search

28 rooms available2026-07-10 → 2026-07-13

HOTEL	ROOM	TYPE	CAPACITY	£/NIGHT (OFF-PEAK)	£/NIGHT (PEAK)	
Verdan London London, UK	1S01	STANDARD DOUBLE	2	£120	£180	View / Book
Verdan London London, UK	1S02	STANDARD DOUBLE	2	£120	£180	View / Book
Verdan London London, UK	1S03	STANDARD DOUBLE	2	£120	£180	View / Book
Verdan London London, UK	1S04	STANDARD DOUBLE	2	£120	£180	View / Book
Verdan Paris Paris, France	2S01	STANDARD DOUBLE	2	£120	£180	View / Book
Verdan Paris Paris, France	2S02	STANDARD DOUBLE	2	£120	£180	View / Book

The search uses real availability logic: it queries SQLite for any overlapping bookings on those dates and excludes them from the results. Because the dates fall in July (peak season), the displayed nightly rates use the pricePeak column rather than priceOffPeak.

Pick any room. On the room detail page, tick breakfast and airport transfer

← Verdan London

Room 1S01

STANDARD DOUBLE · London, UK

ROOM INFORMATION

Type	STANDARD DOUBLE
Capacity	2 guests
Off-peak rate	£120/night
Peak rate	£180/night

Peak season: Jun–Aug, Dec 20–Jan 5.

BOOK THIS ROOM

Check-in

05/18/2026



Check-out

05/19/2026



Optional services

- ☒ Airport Transfer (one-way) — £50
- ☒ Full English Breakfast (per person/day) — £20
- ☐ Spa Access (per person/day) — £35
- ☐ Late Check-out until 2PM — £40

Room (1 night) £120.00
Services £70.00

Total **£190.00**

Confirm Booking

The booking form calculates the total in real time as you tick services or change dates. Service prices come from a fixed catalogue (Breakfast £20, Airport Transfer £50), and the room rate is correctly resolved to the peak figure for July.

Click confirm booking

← Verdan London

Room 1S01

STANDARD DOUBLE · London, UK

ROOM INFORMATION

Type	STANDARD DOUBLE
Capacity	2 guests
Off-peak rate	£120/night
Peak rate	£180/night

Peak season: Jun–Aug, Dec 20–Jan 5.

BOOKING CONFIRMED

Booking confirmed!
Your reservation has been created.

View booking

All my bookings

The booking POST passes through three protections: session auth, room existence + status check, and a date-overlap conflict query. On success the server creates the Booking, BookingRoom, and any BookingService rows inside a single transaction and writes a BOOKING_CREATE entry to the audit log.

My Bookings

[+ New booking](#)[Upcoming \(3\)](#) [Past \(1\)](#)

#	HOTEL	ROOM(S)	CHECK-IN	CHECK-OUT	STATUS	TOTAL			
3	Verdan New York	3P01	20 Jul 2026	25 Jul 2026	CONFIRMED	£3750.00			
11	Verdan London	1S01	10 Jul 2026	13 Aug 2026	CONFIRMED	£6190.00			
2	Verdan Paris	2D01	25 May 2026	28 May 2026	CONFIRMED	£600.00			

The bookings list is scoped to the logged-in guest via the session, so other guests' reservations are never visible. Each row exposes three actions: view details, view invoice, and cancel (only when the booking is still CONFIRMED).

Click the Eye icon to view booking details, then click invoice page

[← Back](#)[Print](#)

Invoice

Booking #3

Verdan New York

New York, USA

UNPAID

BILL TO

John Smith
john.smith@gmail.com

STAY

7/20/2026 → 7/25/2026
5 nights

Rooms

Room	Type	£/night	Nights	Subtotal
3P01	PENTHOUSE	£750	5	£3750

Rooms subtotal	£3750
Services subtotal	£0

Total **£3750**

The invoice page assembles the bill server-side from the priceAtBooking snapshot stored on each BookingRoom. This means the invoice total stays correct even if a manager later changes the nightly rate, satisfying the spec's "fair-pricing" requirement.

Test Case 2 — Staff operations: check-in, services, check-out

This walkthrough proves the front-desk daily workflow: check guests in on arrival, attach extras during the stay, check them out, and settle the invoice.

Log out, then log in as `emma.wilson@verdan.com / Staff123!` (a Verdan London staff member).

The screenshot shows the Verdan Hotels dashboard for a staff member named Emma Wilson. The dashboard is scoped to Hotel #1. It displays the following sections:

- Dashboard:** Today's arrivals, departures, and live occupancy. Includes a Refresh button.
- OCCUPANCY:** A table showing room statuses.
- TODAY'S ARRIVALS (1):** A table with one arrival for Carlos Hernandez in room 1S01, with a Check in button.
- TODAY'S DEPARTURES (1):** A table with one departure for Tom O'Brien in room 1S01, with a Check out button.
- RECENT ACTIVITY:** A list of recent actions including bookings and check-ins.

OCCUPANCY					
TOTAL	OCCUPIED	AVAILABLE	CLEANING	OUT OF SERVICE	OCCUPANCY
10	2	7	0	1	20%

TODAY'S ARRIVALS (1)			
BOOKING	GUEST	ROOMS	
#5	Carlos Hernandez	1S01	<button>Check in</button>

TODAY'S DEPARTURES (1)			
BOOKING	GUEST	ROOMS	
#10	Tom O'Brien	1S01	<button>Check out</button>

RECENT ACTIVITY			
BOOKING_CREATE	Booking #11 (1 room(s), 2026-07-10 -- 2026-08-13)	by John Smith	5/16/2026, 11:28 AM
BOOKING_CREATE	Booking #8 (FAMILY_SUITE)	by Aisha Patel	5/16/2026, 11:59:03 PM
BOOKING_CREATE	Booking #3 (PENTHOUSE)	by John Smith	5/16/2026, 11:59:03 PM
BOOKING_CREATE	Booking #2 (DELUXE_KING)	by John Smith	5/16/2026, 12:00:00 AM
CHECK_IN	Booking #4	by Emma Wilson	5/16/2026, 12:00:00 AM
ROOM_STATUS_CHANGE	Room #1D03: AVAILABLE --> OUT_OF_SERVICE	by Emma Wilson	5/17/2026, 12:59:03 AM
CHECK_IN	Booking #10	by Emma Wilson	5/17/2026, 12:00:00 AM
BOOKING_CREATE	Booking #5 (STANDARD_TWIN)	by Carlos Hernandez	5/17/2026, 12:00:00 AM

The dashboard is scoped to the staff member's assigned hotel (Verdan London) via the hotelId stored in the session. The occupancy row aggregates room statuses live from the database, and Today's Arrivals/Departures tables are populated from the seed.

In Today's arrivals, click Check in next to Carlos Hernandez. On the check-in page, click Confirm check-in.

Check in

BOOKING #5

Guest	Carlos Hernandez carlos.hernandez@gmail.com
Check-in	5/19/2026
Check-out	5/21/2026
Status	CHECKED IN

ROOMS

ROOM	TYPE	NIGHTLY
1501	STANDARD DOUBLE	£120

Check-in complete
The guest has been checked in successfully.

Check-in runs inside a Prisma transaction that updates the booking status to CHECKED_IN, sets every assigned room to OCCUPIED, and writes an audit-log entry. The handler also enforces that the booking's hotel matches the staff member's hotel — Verdan Paris staff cannot check in a Verdan London booking.

Open `/staff/services`. Select Carlos's booking from the dropdown, choose Spa Access, quantity = 2, and click Add.

Ancillary services

Attach extras (breakfast, transfer, spa, late checkout) to active bookings.

Service added.

SERVICE CATALOG

SERVICE	UNIT	PRICE
Airport Transfer	per trip	£50
Breakfast	per person, per day	£20
Spa Access	per person, per day	£35
Late Checkout (until 2 PM)	flat	£40

ADD TO BOOKING

Booking	Service	Qty	
#5 — Carlos Hernandez (CHECKED_IN) ▾	Airport Transfer — £50 (per trip) ▾	1	<button>Add</button>

SERVICES ON BOOKING #5

SERVICE	QTY	PRICE EA.	SUBTOTAL	
AIRPORT TRANSFER	1	£50	£50.00	
AIRPORT TRANSFER	1	£50	£50.00	

Services can be attached to any CONFIRMED or CHECKED_IN booking. They're priced from a fixed catalogue and stored individually so the invoice can break them out by line. A SERVICE_ADD audit log is written, and the same panel allows staff to remove the service again via the trash icon.

Open /staff/rooms. Find any Available room click its CLEANING transition button.

Rooms

Update room status as you clean and inspect.

Status	Type
All ▾	All ▾

ROOM	HOTEL	TYPE	CAPACITY	STATUS	CHANGE STATUS TO
1001	Verdan London	DELUXE KING	2	CLEANING	→ AVAILABLE → OUT_OF_SERVICE
1002	Verdan London	DELUXE KING	2	AVAILABLE	→ CLEANING → OUT_OF_SERVICE
1003	Verdan London	DELUXE KING	2	OUT_OF_SERVICE	→ AVAILABLE
1F01	Verdan London	FAMILY SUITE	4	OCCUPIED	→ CLEANING
1F02	Verdan London	FAMILY SUITE	4	AVAILABLE	→ CLEANING → OUT_OF_SERVICE
1P01	Verdan London	PENTHOUSE	4	AVAILABLE	→ CLEANING → OUT_OF_SERVICE
1501	Verdan London	STANDARD DOUBLE	2	OCCUPIED	→ CLEANING
1502	Verdan London	STANDARD DOUBLE	2	AVAILABLE	→ CLEANING → OUT_OF_SERVICE
1503	Verdan London	STANDARD DOUBLE	2	AVAILABLE	→ CLEANING → OUT_OF_SERVICE
1504	Verdan London	STANDARD DOUBLE	2	AVAILABLE	→ CLEANING → OUT_OF_SERVICE

Back at /staff/dashboard, in Today's departures click Check out on Tom O'Brien. Review the auto-generated invoice preview, then click Process check-out, and finally Mark paid.

Check out

BOOKING #10

CHECKED OUT

Guest	Tom O'Brien tom.obrien@gmail.com
Dates	5/17/2026 → 5/19/2026 (2 nights)

ROOMS

ROOM	TYPE	£/NIGHT	NIGHTS	SUBTOTAL
1501	STANDARD DOUBLE	£120	2	£240

Rooms subtotal	£240
Services subtotal	£0
Total	£240

Check-out complete
Invoice #5 generated for £240

PAID Done

Test Case 3 — Admin management & audit trail

This walkthrough proves system-wide management capabilities: CRUD on hotels, rooms, and users, plus the cross-hotel audit log.

Log out, then log in as sarah.mitchell@verdan.com / Admin123!

Open admin/users.

Verdan

Dashboard
Bookings
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Hotels
All Rooms
Audit Logs

Logout

Verdan Hotels

Sarah Mitchell **ADMIN** [Logout](#)

Users

All user accounts across roles. Click a name to edit.

All roles

NAME	EMAIL	ROLE	STATUS	
Tom O'Brien	tom.obrien@gmail.com	GUEST	Active	GUEST
John Smith	john.smith@gmail.com	GUEST	Active	GUEST
Linda Johnson	linda.johnson@gmail.com	GUEST	Active	GUEST
Carlos Hernandez	carlos.hernandez@gmail.com	GUEST	Active	GUEST
Aisha Patel	aisha.patel@gmail.com	GUEST	Active	GUEST
Michael Garcia	michael.garcia@verdan.com	STAFF	Active	STAFF
Olivia Williams	olivia.williams@verdan.com	STAFF	Active	STAFF
Emma Wilson	emma.wilson@verdan.com	STAFF	Active	STAFF
David Brown	david.brown@verdan.com	STAFF	Active	STAFF
Sophia Taylor	sophia.taylor@verdan.com	STAFF	Active	STAFF
Lucas Martin	lucas.martin@verdan.com	STAFF	Active	STAFF
Camille Leroy	camille.leroy@verdan.com	STAFF	Active	STAFF

130 rows

Admins see every account in the system (currently 18 users: 1 admin, 3 managers, 9 staff and 5 guests). Role badges colour-code each account, the inline dropdown lets admin change a user's role immediately, and locked accounts surface a red badge for one-click unlocking.

Open admin/hotels and click Add hotel. Fill in Verdan Tokyo / Tokyo / Japan and save.

Verdan

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Logout

Verdan Hotels

Sarah Mitchell **ADMIN** [Logout](#)

Hotels

Manage Verdan properties and their basic details.

[+ Add hotel](#)

NAME	CITY	COUNTRY	ROOMS	
Verdan London	London	UK	10	Edit
Verdan New York	New York	USA	10	Edit
Verdan Paris	Paris	France	10	Edit
Verdan Tokyo	Tokyo	Japan	0	Edit

New hotel added.

Open admin/rooms. Click Edit on any row and change the capacity and click Save changes.

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Rooms

Edit room

ROOM 1D02 · VERDAN LONDON

Room number

1D02

Capacity

4

Type

DELUXE KING

Off-peak price (£)

180

Peak price (£)

250

Status

AVAILABLE

Save changes

he room edit page supports the full set of fields — room number, type, capacity, peak/off-peak prices, status — with field validation at both the form and API levels. The change is recorded as ROOM_UPDATE in the audit log.

Back on admin/rooms, find the room that Linda Johnson is currently checked in to (her `FAMILY\_SUITE` at Verdan London) and click its trash icon. Confirm the deletion.

Verdan

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Rooms

All rooms across every Verdan property.

Hotel: All hotels + Add room

Cannot delete room — 1 active booking(s) reference it

ROOM	HOTEL	TYPE	CAPACITY	OFF-PEAK £	PEAK £	STATUS	
1D01	Verdan London	DELUXE KING	2	£180	£250	CLEANING	Edit
1D02	Verdan London	DELUXE KING	2	£180	£250	AVAILABLE	Edit
1D03	Verdan London	DELUXE KING	2	£180	£250	OUT OF SERVICE	Edit
1F01	Verdan London	FAMILY SUITE	4	£240	£320	OCCUPIED	Edit
1F02	Verdan London	FAMILY SUITE	4	£240	£320	AVAILABLE	Edit
1P01	Verdan London	PENTHOUSE	4	£500	£750	AVAILABLE	Edit
1S01	Verdan London	STANDARD DOUBLE	2	£120	£180	CLEANING	Edit
1S02	Verdan London	STANDARD DOUBLE	2	£120	£180	AVAILABLE	Edit
1S03	Verdan London	STANDARD DOUBLE	2	£120	£180	AVAILABLE	Edit
1S04	Verdan London	STANDARD DOUBLE	2	£120	£180	AVAILABLE	Edit
2D01	Verdan Paris	DELUXE KING	2	£180	£250	AVAILABLE	Edit
2D02	Verdan Paris	DELUXE KING	2	£180	£250	AVAILABLE	Edit
2D03	Verdan Paris	DELUXE KING	2	£180	£250	AVAILABLE	Edit

The DELETE handler refuses to remove a room while any active bookings (CONFIRMED or CHECKED_IN) reference it. This protects referential integrity — historical CHECKED_OUT bookings, however, keep their priceAtBooking snapshot and don't block the delete, so old data is preserved.

Open admin/audit-logs

Verdan

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Verdan Hotels

Sarah MitchellADMINLogout

Audit logs

Filter by action and date range. API_ACCESS rows show authenticated requests.

Action

From

To

Refresh

WHEN	ACTION	USER	DETAILS	IP
5/19/2026, 1:38:15 AM	API_ACCESS	Sarah Mitchell sarah.mitchell@verdan.com	GET /api/audit-logs?page=1&pageSize=50	111
5/19/2026, 1:36:45 AM	API_ACCESS	Sarah Mitchell sarah.mitchell@verdan.com	DELETE /api/admin/rooms/8	111
5/19/2026, 1:36:17 AM	API_ACCESS	Sarah Mitchell sarah.mitchell@verdan.com	GET /api/staff/rooms	111
5/19/2026, 1:34:24 AM	API_ACCESS	Sarah Mitchell sarah.mitchell@verdan.com	GET /api/staff/rooms	111
5/19/2026, 1:33:08 AM	HOTEL_CREATE	Sarah Mitchell sarah.mitchell@verdan.com	Hotel #4 (Verdan Tokyo)	—
5/19/2026, 1:33:08 AM	API_ACCESS	Sarah Mitchell sarah.mitchell@verdan.com	POST /api/admin/hotels	111
5/19/2026, 1:31:04 AM	API_ACCESS	Sarah Mitchell sarah.mitchell@verdan.com	GET /api/admin/users	111
5/19/2026, 1:31:01 AM	API_ACCESS	Sarah Mitchell sarah.mitchell@verdan.com	GET /api/staff/dashboard	111
5/19/2026, 1:31:01 AM	API_ACCESS	Sarah Mitchell sarah.mitchell@verdan.com	GET /api/staff/dashboard	111
5/19/2026, 1:31:01 AM	API_ACCESS	Sarah Mitchell sarah.mitchell@verdan.com	GET /api/auth/session	111
5/19/2026, 1:31:01 AM	LOGIN	Sarah Mitchell sarah.mitchell@verdan.com	—	111

203 ms

The audit log records

> CHECK_IN, CHECK_OUT, INVOICE_PAID, ROOM_STATUS_CHANGE and ROOM_RATE_CHANGE. Each row captures the user, action, details, IP address, and timestamp.